

A two-stage learning architecture for reliable financial time series forecasting

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ABSTRACT

Financial time series are noisy, non-stationary, and volatile, which makes accurate forecasting particularly challenging. State-of-the-art time series foundation models achieve strong performance on benchmark problems with clear seasonality or trend, such as energy demand or traffic forecasting, yet struggle to produce reliable forecasts in low signal-to-noise environments such as financial markets. To address this limitation, we introduce a two-stage learning architecture that reformulates short-term trend forecasting as a selective classification problem. The first stage consists of a primary predictive model that outputs directional forecasts (up/down), while the second stage introduces a learned reliability estimator that models the error characteristics of the primary model. This explicit decoupling of prediction and reliability estimation enables the system to learn when the primary model is likely to produce an incorrect forecast. The proposed architecture supports selective inference, where decisions are executed only for instances classified as reliable by the second stage. We formalize this mechanism within a risk-coverage optimization framework, which provides explicit control over the trade-off between prediction precision and execution frequency. The proposed approach is model-agnostic and can be seamlessly integrated with modern neural forecasting models, including time series foundation models. Extensive experiments on cryptocurrencies and equity markets demonstrate consistent improvements in precision (+14% on average) and risk reduction (−14%) compared to non-selective baselines. Historical backtests with a simple trading strategy confirm the practical viability of the approach.

1. Introduction

Forecasting in finance has long been a primary objective for financial researchers and investors. However, accurately predicting the prices of assets such as stocks or cryptocurrencies remains difficult due to non-stationarity, low signal-to-noise ratio, and extreme volatility [14].

Recently, the tremendous success of Transformer-based architectures [53] has been witnessed in Natural Language Processing (NLP) and Computer Vision, especially exemplified by models like the GPT family [41] and the Vision Transformer (ViT) [17]. A key feature of these models is their ability to perform as few-shot or even zero-shot reasoners, requiring minimal fine-tuning [9]. Inspired by these capabilities, researchers have begun to formulate time series forecasting as a next-token prediction task, analogous to language modeling. This has led to

the development of dedicated Time Series Foundation Models trained on massive temporal datasets [3,5,30,44].

However, a critical **limitation** remains. While these models excel at capturing clear periodicities, they usually struggle with the specific nature of financial data.

Recent studies have shown that generic LLMs fail when data lack clear patterns [51], and even specialized foundation models struggle with the irregularity of financial time series [36].

To address this challenge, we propose enhancing Time Series Foundation Models for finance by decoupling the forecasting problem into two distinct inquiries:

1. Directional Prediction: What is the probable movement of the asset price (up or down)?

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2. Prediction Reliability: Is the directional forecast reliable or not?

This kind of decision decoupling was first introduced in [14] as the meta-labeling concept. Here, a primary model $M1$ predicts the direction of the price action (up or down), while a distinct secondary model $M2$ estimates the confidence or reliability of $M1$'s prediction.¹

This decoupling offers three key advantages. First, it mitigates overfitting via decomposition, ensuring that the distinct tasks of directional prediction and confidence estimation do not confound one another. Second, it addresses market asymmetry, allowing the model to treat the reliability of 'Up' and 'Down' signals independently to account for regime-specific dynamics. Third, it shifts the optimization focus to Precision and Risk. The secondary model acts as a filter, prioritizing the elimination of False Positives over the inclusion of marginal opportunities.

This approach is critical in high-stakes environments such as healthcare, autonomous driving, and algorithmic trading [48], where there is little tolerance for False Positives [22,43]. In a trading context, a False Positive occurs when the model predicts an asset price increase (decrease), but the market moves continuously downward (upward). Such errors can lead to financial losses and should be minimized.

By filtering out unreliable predictions, we move towards selective classification [20,22]. This technique introduces a trade-off: it significantly improves predictive precision (low risk) but at the cost of reduced coverage (the percentage of samples for which the model provides a prediction).

The core **challenge** is therefore to design a mechanism that maximizes predictive performance on the selected data points while maintaining sufficient coverage to capture trading opportunities. To this end we propose to use a two-stage learning architecture. Specifically, we utilize several state-of-the-art Time Series Foundation Models as our primary model ($M1$), including Chronos and Chronos-Bolt [3], TimeGPT-1 [21], and TiRex [5]. For the secondary model ($M2$), we design a hybrid architecture combining Convolutional Neural Networks (CNNs) and Transformers, inspired by Zeng et al. [59]. This hybrid approach is ideal for financial data: CNNs effectively capture local patterns (short-term price action), while the Transformer's self-attention mechanism models long-range dependencies and seasonality [64]. We evaluate our two-stage learning architecture across diverse assets including equities and cryptocurrencies, based on daily price movements.

The core **contributions** of this paper are as follows:

- We formalize short-term financial trend forecasting as a selective inference task within a two-stage architecture where a primary model predicts the direction of price movement (up, down), while a secondary model estimates the reliability of the primary model by learning which input sequences are associated with reliable versus unreliable predictions. The resulting two-stage learning architecture enables selective decision-making, where predictions are executed only when deemed sufficiently reliable.
- We maximize predictive performance by formulating the decision process as a risk-coverage optimization problem, which provides explicit control over the trade-off between the precision of a prediction and execution frequency.
- We demonstrate improved precision and economic viability across multiple cryptocurrencies and equities.

The remainder of the paper is organized as follows. [Section 2](#) reviews the related literature. [Section 3](#) introduces the notation and key definitions, and formally states the problem of selective financial time series short-term trend forecasting. [Section 4](#) describes our methodology. [Section 5](#) details the experimental design. [Section 6](#) presents empirical results, ablation studies, and backtesting, together with a

discussion of limitations. Finally, [Section 7](#) concludes the paper and outlines promising avenues for future work.

2. Related work

This section surveys the literature related to our work. The discussion is structured around three key areas: selective prediction, Time Series Foundation Models, and the application of deep learning and large language models in finance.

2.1. Selective prediction

Selective classification, also called prediction with a "reject option", aims to improve system reliability by allowing models to abstain when classification confidence is low. Early approaches [22] proposed single-model architectures that output both a prediction and a selection score. This was refined in SelectiveNet [23], an end-to-end deep neural network that optimizes a joint loss for prediction and rejection. Similarly, Deep Gamblers [37] reframes the problem as a classification task with an additional "abstain" class, training the model using a portfolio-theoretic loss function. However, these end-to-end approaches have two limitations relevant to our study: (1) they are primarily designed for standard classification rather than complex time series forecasting, and (2) they entangle the forecasting and reliability tasks. A comprehensive review of these methods is provided in [24].

More recent work focuses on training dynamics to identify uncertainty. For example, Rabanser et al. [43] track the volatility of predictions across training epochs; instances with high disagreement between intermediate and final outputs are deemed unreliable. Huang et al. [25] propose a self-adaptive training scheme that exponentially averages predictions to correct noisy labels. [42] uses a Deep Q-Network to learn trading timing generate action-based labels and also applies instance selection to denoise the training set before training a supervised model.

Conformal Prediction (CP) offers a statistically rigorous framework for uncertainty quantification, generating prediction sets with guaranteed coverage [2,46]. However, standard CP assumes data exchangeability: data points can be permuted without changing the joint distribution. This assumption is fundamentally violated in time series data, where temporal order is critical. While recent adaptations attempt to address non-exchangeability [57], their applicability to multi-step forecasting remains limited. Beyond CP, Wong et al. [54] show that standard neural network uncertainty estimates are miscalibrated under the volatility clustering typical of financial data, further motivating the need for dedicated reliability estimation mechanisms.

In contrast to the methods above, the meta-labeling concept suggests separating forecasting from reliability estimation [14]. [28] demonstrated its application to explainable financial forecasting. Our work builds upon this paradigm.

2.2. Time series foundation models (TSFMs)

The success of Large Language Models (LLMs) has catalyzed the development of foundation models for time series analysis. Chronos [3] is a leading univariate model that tokenizes time series and processes them via a Transformer architecture, pre-trained on approximately 3 billion context-target segments. Its successor, ChronosX [4], adds capabilities for exogenous variables, though it is not yet available for fine-tuning, a key requirement for our methodology. TimeGPT-1 [21] was among the first general-purpose models, designed primarily for zero-shot forecasting across heterogeneous domains. Other notable models include TimesFM [13], Moirai [55], and Tiny Time Mixers [18]. Recently, TiRex [5] has demonstrated strong zero-shot performance leveraging the xLSTM architecture [7].

However, a critical limitation remains: these general-purpose TSFMs are not explicitly trained or designed for the high noise and non-stationarity characteristics of financial data. To address this, we select

¹ While [14] originally framed this as determining the *size* of a bet based on the *side*, our work focuses on forecasting reliability rather than position sizing.

Chronos, Chronos-Bolt, TimeGPT-1, and TiRex to serve as our primary models. These selections represent the state-of-the-art in both fine-tunable and zero-shot time series forecasting.

2.3. Deep learning and LLMs in finance

Over the last decade, Convolutional Neural Networks (CNNs) and Transformers have become popular for financial time series analysis. [8,39,52] provide extensive studies of machine learning and deep learning techniques applied to financial market prediction, while [38,49,62] review the application of attention mechanisms in quantitative trading.

Hybrid attention-based architectures have been applied to portfolio management via reinforcement learning [45] and to stock prediction on the S&P 500 and NASDAQ using multi-head cross-attention with GRU [65]. Echo State Networks, a variant of Recurrent Neural Networks, have also been applied to predict financial time series [47].

Semi-supervised deep learning methods have been applied to Bitcoin anomaly detection [56]. Furthermore, Chen et al. [12] use a CNN-based framework to quantify financial investment ensuring investment returns and reducing investment risks. Jing et al. [31] employ a hybrid CNN-LSTM approach in which a CNN classifies investor sentiment and an LSTM predicts stock prices. Of particular relevance to our secondary model is the CTTS architecture [59], which combines CNNs for local feature extraction with Transformers for global temporal modeling. Hybrid approaches using CNNs have been successfully employed for price direction prediction [15] and chart pattern recognition of financial time series [16].

Beyond model-level innovations, a complementary line of work explores meta-learning paradigms for rapid adaptation across financial markets. Noor and Fatima [40] propose an XGBoost-based meta-learning framework with dynamic feature engineering, adjusting indicators such as RSI and Bollinger Band Width to evolving market conditions, pre-trained on multiple equities and cryptocurrencies and fine-tuned on new instruments with minimal retraining. Batool et al. [6] further develop a Meta-SGD formulation that jointly forecasts equity (S&P 500, NASDAQ) and cryptocurrency (Bitcoin, Ethereum) indices, reporting gains over standalone deep learning baselines under limited data and computational budgets. While these works share the broader “learning to learn” philosophy, they target cross-market generalization, which is orthogonal to the per-forecast reliability estimation pursued in this work.

Beyond standard deep learning, researchers have begun adapting LLMs for finance. [50] analyze financial markets by incorporating LLM-inferred social context features into stock forecasting. They construct a chain of daily posts, which LLMs summarize and analyze to derive sentiment scores and a buy/sell indicator; these features are then combined with historical financial data and passed to a forecasting module based on deep learning models (e.g., LSTMs, TCNs, Transformers, and GANs) to predict stock movements.

Time-LLM [30] reprograms time series into text-like embeddings to leverage pre-trained language models. Chaffard et al. [11] recently improved Time-LLM for Bitcoin forecasting by integrating technical indicators and sentiment analysis via prompt engineering. While these generative approaches show promise [32,35,58], assessing their reliability in high-stakes environments remains a challenge, motivating our use of a dedicated learned reliability layer.

For completeness, we note an adjacent but methodologically distinct line of research that investigates adaptive control for stochastic non-linear dynamical systems, with applications including flexible robotic manipulators [61], DC-DC power converters [63], and multi-agent ship maneuvering [60]. These works address continuous-time tracking and stabilization under Brownian-noise disturbances using tools such as adaptive fuzzy control, command-filtered backstepping, and finite/predefined-time stability, a methodological tradition distinct from the discrete-time selective classification framework pursued here.

3. Preliminaries and problem statement

In this section, we introduce the necessary notations relevant to the framework, followed by a formal definition of the *selective financial time series trend forecasting* problem.

3.1. Preliminaries

We first establish the definitions for the financial time series structure, the labeling mechanism, and the metrics for selective classification in finance.

Definition 3.1 (Univariate Financial Time Series). We consider a univariate time series of closing prices $P = \{p_t\}_{t=1}^T$ where $p_t \in \mathbb{R}_{>0}$ represents the closing price of an asset at the time step t .

Definition 3.2 (Lookback and Horizon). For a given time step t , the input to our model is a history window (lookback) of length L . We denote this sequence as $X_t = \{p_{t-L+1}, \dots, p_t\}$, a subsequence of P with length L . In this study, we set the lookback window to $L = 90$ days. The goal is to forecast the short-term trend over a horizon $h = 7$ days.

The lookback $L = 90$ days covers roughly one trading quarter, providing the model with sufficient recent history to capture prevailing volatility regimes without diluting the signal with stale data. The horizon $h = 7$ days targets a swing-trading approach, which is sufficiently short for learning directional signals, and long enough to filter out intraday noise. Both windows slide forward by one day at each time step.

Definition 3.3 (Ground Truth Short-term Trend). We define the future return $r_{t,h}$ as the relative price change between time t and $t + h$:

$$r_{t,h} = \frac{p_{t+h} - p_t}{p_t} \quad (1)$$

The ground truth short-term trend label is ternary, $y_t \in \{\text{Up}, \text{Down}, \text{Same}\}$, and is determined by a threshold ρ that filters out insignificant price movements:

$$y_t = \begin{cases} \text{Up} & \text{if } r_{t,h} > \rho \\ \text{Down} & \text{if } r_{t,h} < -\rho \\ \text{Same} & \text{if } -\rho \leq r_{t,h} \leq \rho \end{cases} \quad (2)$$

Across all assets, we set the movement threshold to $\rho = 0.2\%$, chosen to cover round-trip transaction fees so that only economically meaningful directional moves are labeled as Up or Down.

At inference, the primary model outputs a ternary forecast $\hat{y}_t \in \{\text{Up}, \text{Down}, \text{Same}\}$. *Same* is assigned when neither predicted directional move exceeds its decision threshold ρ (details in Section 4). At a minimum, ρ should be set above the effective trading-cost level so that directional signals correspond to economically meaningful moves. Because the secondary task only concerns forecasts that take a side, we evaluate the directional task on the index set of samples for which $M1$ predicts a direction:

$$\mathcal{I}_{\text{dir}} = \{t : \hat{y}_t \in \{\text{Up}, \text{Down}\}\}. \quad (3)$$

Definition 3.4 (Meta Labeling). Following the concept in [14], let $\hat{y}_t$ be the prediction made by a primary model $M1$. On the samples where $M1$ predicts a direction ($t \in \mathcal{I}_{\text{dir}}$), the meta-label $z_t \in \{0, 1\}$ indicates whether that directional prediction was correct:

$$z_t = \mathbb{1}(\hat{y}_t = y_t) \quad (4)$$

where $\mathbb{1}(\cdot)$ is the indicator function. The secondary model $M2$ is trained to predict z_t , effectively estimating the reliability of $M1$'s results.

At inference time, $M2$ outputs a reliability score $\pi_t \in [0, 1]$ estimating $P(z_t = 1 \mid X_t)$. The predicted meta-label is then obtained by thresholding this score,

$$\hat{z}_t = \mathbb{1}(\pi_t \geq \theta), \quad \hat{z}_t \in \{0, 1\}. \quad (5)$$

3.2. Coverage and risk

In a standard classification task, the model must provide a prediction for every sample. In our two-stage learning architecture, the system can abstain. We report selective metrics on directional samples only ($\mathcal{I}_{\text{dir}}$). Define the executed directional set as

$$\mathcal{E} = \{t \in \mathcal{I}_{\text{dir}} : \hat{z}_t = 1\} \quad (6)$$

We therefore evaluate performance using two specific metrics: Coverage and Risk.

Definition 3.5 (Coverage). Coverage is defined as the proportion of samples for which the secondary model $M2$ grants approval (i.e., decides to execute). High coverage implies the system provides many tradable opportunities. Formally:

$$\text{Coverage} = \frac{|\mathcal{E}|}{|\mathcal{I}_{\text{dir}}|}. \quad (7)$$

In this paper, the terms ‘coverage’ and ‘execution rate’ are used interchangeably.

Definition 3.6 (Risk). Risk is the error rate only on executed directional samples:

$$\text{Risk} = \frac{1}{|\mathcal{E}|} \sum_{t \in \mathcal{E}} \mathbb{1}(\hat{y}_t \neq y_t). \quad (8)$$

3.3. Problem definition

We can now state the core objective of this paper. The primary task is binary directional (Up, Down) trend forecasting over a fixed horizon. The proposed secondary (selective) task is to determine, on a per-forecast basis, whether a directional prediction should be acted upon or abstained from.

Problem 1. [Selective Financial Time Series Short-term Trend Forecasting] Given a financial time series dataset, the objective is to train a primary model $M1$ and a secondary model $M2$, then choose side-specific thresholds by maximizing coverage under a risk cap α , while maintaining minimum practical coverage. Specifically, we seek configurations that achieve lower risk than the standalone primary model in this selective execution setting.

4. Methodology

In this section, we detail our approach for selective financial time series forecasting. We first introduce the overall two-stage learning architecture, followed by the primary model ($M1$) for time series forecasting and the secondary model ($M2$) for reliability estimation. Finally, we outline the optimization procedure for balancing risk and coverage.

4.1. Two-stage learning architecture

Fig. 1 illustrates the high-level workflow of the architecture. Unlike traditional forecasting approaches that focus solely on the prediction at every time step, this framework introduces a reliability layer. The architecture comprises three distinct stages (see also Algorithm 1 in Appendix A):

- **Short-term Trend Forecasting Model ($M1$).** A time series foundation model generates a directional forecast for the target horizon (e.g., $t + 7$). During the training phase, correctness is evaluated on the samples where $M1$ predicts a direction (i.e., excluding predicted *Same*) to generate binary meta labels: 1 for correct, 0 for incorrect (see Definition 3.4).

- **Reliability Estimation Model ($M2$).** A secondary classifier is trained using the input series X_t and the meta labels generated by $M1$. The objective of $M2$ is to distinguish between reliable and unreliable forecasts made by $M1$.
- **Selective Execution ($M1 + M2$).** In the inference phase, a prediction (trade) is only ‘executed’ if two conditions are met: (1) $M1$ predicts a specific direction (Up or Down) with sufficient confidence, and (2) $M2$ deems this prediction reliable.

Note that we explicitly differentiate between Up-trend and Down-trend models, allowing the system to learn the distinct characteristics associated with rising versus falling markets.

4.2. Primary model: time series foundation models

We employ a diverse set of state-of-the-art general purpose time series foundation models as our primary models: Chronos and Chronos-Bolt [3], TimeGPT-1 [21], and TiRex [5]. These models process the historical lookback window X_t and output a regression forecast for the prediction horizon. Then we convert these regression outputs into the classification labels required for the two-stage learning architecture based on the threshold ρ .

In particular, these foundation models can be broadly categorized into two types: Probabilistic models (Chronos, Chronos-Bolt) are fine-tuned on our financial datasets. Since they are non-deterministic, we generate 30 independent forecast paths for each prediction. We calculate an empirical “direction probability” based on the fraction of paths that cross the threshold ρ . We then determine an optimal probability threshold τ_{prob} using the Receiver Operating Characteristic (ROC) curve on the validation set, balancing True Positives and False Positives.

TimeGPT-1 and TiRex are used in a zero-shot fashion: TimeGPT-1 via Nixtla’s hosted API and TiRex via its publicly released checkpoint. In both cases, the direction is determined strictly by whether the point forecast exceeds ρ .

4.3. Secondary model: reliability estimation models

As a secondary model, we develop a hybrid architecture that combines Convolutional Neural Networks (CNNs) and Transformers, inspired by Zeng et al. [59]. Unlike prior work, which directly forecasts prices, our modified architecture acts as a binary classifier that predicts the *reliability* of $M1$ ’s predictions. This architectural choice is motivated by three considerations:

- **Local feature extraction:** 1D CNN kernels slide over the input time series to capture short-term dependencies and local volatility patterns.
- **Global context modeling:** The CNN output is fed into a Transformer encoder, where the self-attention mechanism enables the model to capture long-range dependencies and seasonality effects that local convolutions alone might miss.
- **Latent representation:** The Transformer output is aggregated into a latent embedding vector, which is passed through a Multilayer Perceptron (MLP) with a softmax activation to produce the final reliability score $\pi_t \in [0, 1]$. This continuous score is then thresholded at θ to yield the binary execute/abstain decision (Definition 3.4, Eq. (5)).

To adapt these considerations specifically for the non-stationary nature of financial data, we propose the following two improvements to [59]. Financial time series suffer from significant distributional shifts (non-stationarity), where statistical properties such as mean and variance change rapidly over time [10,19]. To address this, *first*, we integrate Reversible Instance Normalization (RevIN) [34]. RevIN normalizes the input to remove non-stationary information before the model proceeds, and (conceptually) restores it at the output, allowing the model to focus on learning the underlying structural patterns rather than the scale.

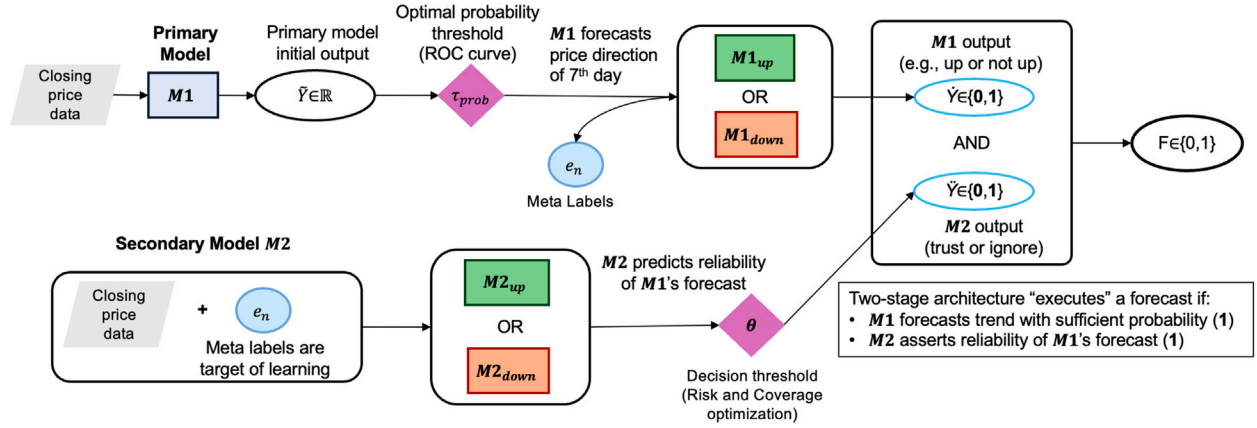


Fig. 1. The proposed two-stage learning architecture in this study. The pipeline decouples the directional prediction ($M1$) from the reliability estimation ($M2$). Separate instances are trained for short-term Up-trend and Down-trend tasks to capture asymmetric market dynamics.

Classic Transformers use fixed sinusoidal positional encoding [53]. Even though fixed encodings can be effective for general-purpose sequence modeling, they cannot adapt to domain-specific temporal structures. Therefore, *second*, we replace the fixed encoding with a learnable positional embedding, initialized from sinusoidal values and updated during training. This preserves the spectral inductive bias of the sinusoidal initialization and allows the model to refine the positional representation to the statistical properties of financial sequences.

4.4. Risk–coverage optimization

Finally, the outputs of the primary model ($M1$) and secondary model ($M2$) are integrated to solve the optimization problem defined in Problem 1. The core challenge is to convert the continuous reliability score output by $M2$ into a discrete binary decision (Execute vs. Abstain). This is controlled by the decision threshold $\theta \in [0, 1]$. By varying θ , we trade off between the frequency of trading (Coverage) and the reliability of those trades (Risk). Next, we formalize this optimization.

As detailed in Section 4.1, we maintain separate reliability models for the Up and Down sides. Consequently, we perform the threshold optimization independently for each side. Let D_{val} denote the validation dataset and s be the direction (side) of the short-term trend. For a specific side s , and a candidate threshold θ , let $Risk_{val}^s(\theta)$ and $Coverage_{val}^s(\theta)$ denote the risk and coverage metrics computed on D_{val} . We enforce a risk cap α (e.g., $\alpha = 20\%$) and a minimum coverage C_{min} (e.g., $C_{min} = 5\%$).

We define the feasible threshold set

$$\Theta_{feas}^s = \left\{ \theta \in [0, 1] : \begin{array}{l} Risk_{val}^s(\theta) \leq \alpha, \\ Coverage_{val}^s(\theta) \geq C_{min} \end{array} \right\}. \quad (9)$$

The optimal threshold θ^s is then chosen by risk-constrained coverage maximization:

$$\theta^s = \arg \max_{\theta \in \Theta_{feas}^s} Coverage_{val}^s(\theta). \quad (10)$$

To solve this optimization problem efficiently and robustly, we employ Optuna [1], an automated hyperparameter optimization framework. Instead of directly searching for θ^s , we define an Optuna study to optimize the model hyperparameters (e.g., learning rate) while the threshold θ^s is determined analytically for each trial.

The optimization process is as follows:

1. **Trial Sampling:** In each trial, Optuna samples a configuration of hyperparameters. The model is trained and validation scores are generated.

2. **Threshold Selection:** On the validation set, we compute the risk-coverage curve and select θ^s that maximizes $Coverage_{val}^s(\theta)$ under the feasibility constraints $Risk_{val}^s(\theta) \leq \alpha$ and $Coverage_{val}^s(\theta) \geq C_{min}$. The unique probability scores predicted by the model on the validation set serve as the candidate grid to locate the threshold θ^s .
3. **Pruning Mechanism:** To enforce feasibility, if no threshold satisfies both constraints, the trial is pruned.
4. **Selection:** The study returns the model hyperparameters that yield the highest valid coverage.

At test time, the optimal threshold θ^s identified by Optuna is frozen and applied to the test set. Performance metrics are reported strictly on the subset of test samples selected by this threshold.

5. Experimental setup

This section details the datasets, experimental configurations, and evaluation metrics.

5.1. Datasets

We evaluate our framework on a diverse set of financial assets, encompassing both equities and cryptocurrencies. For $M1$ fine-tuning, we utilize broader datasets to learn robust feature representations. Specifically, the equity training set includes the top 20 S&P 500 constituents by weight, plus S&P 500 and Nasdaq-100 stock indices, sourced from Yahoo Finance.² Similarly, the cryptocurrency training set comprises the top 20 cryptocurrency assets by market capitalization, sourced from Binance.³ The date ranges for all assets are detailed in Table B.1 in the Appendix B.

We note a modest trend bias and class imbalance in the data. On average, equities demonstrate class imbalance with 56% upward movements, 38% downward, and 6% flat periods, whereas crypto (Bitcoin and Ethereum) exhibits on average 51% upward movements, 45% downward, and 4% flat periods.

For inference time, we report our framework performance on the following equities and cryptocurrencies. Specifically, we focus on:

- Cryptocurrencies: Bitcoin (BTC) and Ethereum (ETH) from Feb 1, 2018 to Nov 11, 2025 (2842 calendar days).
- Stock Indices: S&P 500 (SPY) and Nasdaq-100 (QQQ) from Jan 1, 2000 to Nov 11, 2025 (6506 trading days).

² <https://finance.yahoo.com/>

³ <https://www.binance.com/>

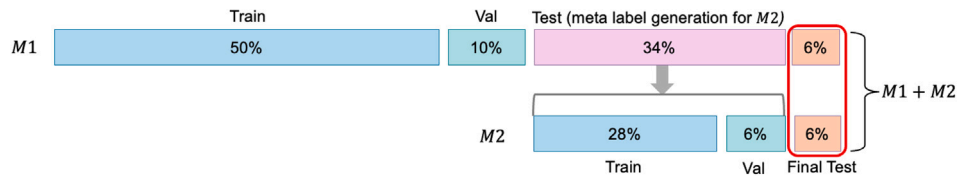


Fig. 2. Chronological data-splitting scheme. *Top*: Initial $M1$ uses a 50/10/40 split. On the final 40% segment the frozen $M1$ generates predictions; of this segment, (A) 34% is used to (i) generate the meta labels for $M2$ and (ii) train and validate $M2$, and (B) the last 6% is held out for testing. *Bottom*: The 40% prediction segment is internally split 70/15/15 (corresponding to 28/6/6 of the full timeline) for $M2$ training, validation, and testing. The final 6% of the timeline (red box) is held out and used to evaluate both the $M1$ -only baseline and the full $M1 + M2$ system under identical conditions.

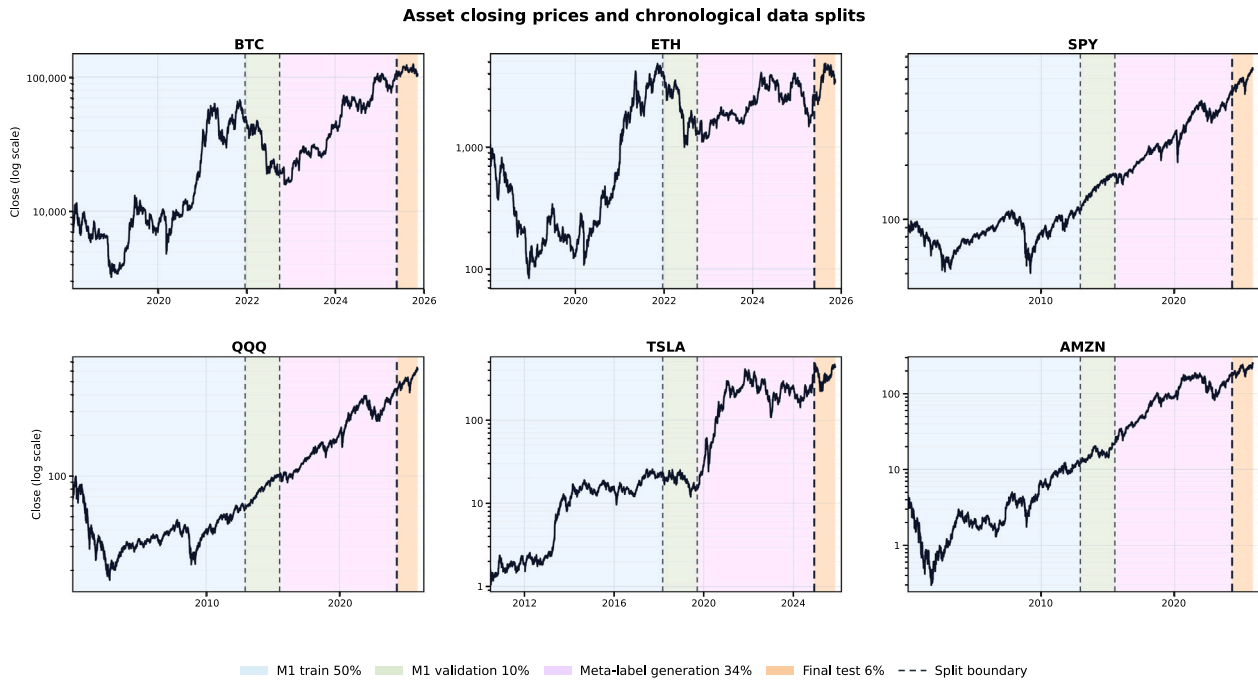


Fig. 3. Closing-price evolution for the six assets used in our experimental protocol, with the chronological train/validation/test splits shaded. Each asset is split into a 50% $M1$ training segment, a 10% $M1$ validation segment, a 34% meta-label generation segment (within which $M2$ is trained on 28% and validated on 6%), and a final 6% held-out test window. The test window (rightmost shaded region) is used to evaluate both the $M1$ -only baseline and the full $M1 + M2$ system under identical conditions. Asset date ranges differ due to differing data availability on Yahoo Finance / Binance.

- Large-Cap Stocks: Tesla Inc. (TSLA) from Jun 29, 2010 to Nov 11, 2025 (3869 trading days), and Amazon.com Inc. (AMZN) from Jan 1, 2000 to Nov 11, 2025 (6506 trading days).

Fig. 2 shows how we construct the training, validation, and test sets for $M1$ only and for $M1 + M2$. Since $M2$ learns from the errors of $M1$, it requires a distinct dataset of meta labels. We therefore employ a two-stage splitting process:

- Stage 1 (Meta Label Generation): We initially split the full timeline into 50% Training, 10% Validation, and 40% prediction segment. We train an initial instance of $M1$ on the 50% segment and generate predictions for the subsequent 40% segment, whereby 34% of these predictions are compared against ground truth to create the meta labels, and the final 6% is held out for testing.
- Stage 2 ($M2$ Training): This 40% segment is further subdivided 70/15/15 (28/6/6 of the full timeline) into $M2$ Training, Validation, and Test; the 28% + 6% train/validation portion is the 34% meta label generation block in Fig. 2, and the $M2$ test set is the final 6% of the data. The 70/15/15 ratio reserves the majority of meta-labeled data for training $M2$, preserves sufficient validation data for side-specific threshold (θ) selection via Optuna (Section 4.4), and holds out a final clean test partition.

- Baseline Comparison: To ensure a fair comparison between the standalone $M1$ forecaster and the full $M1 + M2$ system, both use the same $M1$ instance, which is trained on the initial 50% of the timeline with the 10% validation segment used for hyperparameter selection, and are evaluated on the same final 6% test window. No retraining of $M1$ on additional data is performed for the baseline; the two systems differ only in whether $M2$'s reliability filtering is applied to $M1$'s predictions or not.

All splits are strictly chronological and temporally non-overlapping: the $M1$ training segment precedes the $M1$ validation segment, which precedes the meta label generation segment (within which $M2$ is trained and validated), which in turn precedes the final test window. No data from a later segment is used at any stage of training, validation, or hyperparameter selection for any earlier-stage model. Lookback windows of length $L = 90$ used to construct prediction inputs may include historical observations from earlier segments. This represents the legitimate use of past data to predict the future and does not introduce future-to-past information leakage.

The final 6% test window is held out and is touched only during evaluation, where it is used to assess both the $M1$ -only baseline and the full $M1 + M2$ system under identical conditions. Fig. 3 illustrates the closing-price evolution and chronological data splits for each asset.

Table 1
Search space of the hyperparameters for Optuna.

Hyperparameter	Symbol	Search space
Learning rate	η_{lr}	$[10^{-6}, 10^{-2}]$ (log-scale)
Fine-tuning steps	S_{ft}	$\{2000, 3000, \dots, 25,000\}$
Batch size	B_{ft}	$\{32, 64, 128, 256\}$
Warmup ratio	r_{warmup}	$[10^{-3}, 10^{-1}]$
Weight decay	λ_{wd}	$[10^{-5}, 10^{-1}]$

5.2. Experimental setup

All experiments are conducted on a single NVIDIA RTX A6000 GPU. We implement automatic mixed precision (AMP) to optimize memory usage and computational efficiency. Random seeds are fixed across all runs to ensure reproducibility.

Primary Model Fine-Tuning We fine-tune the probabilistic foundation models (Chronos and Chronos-Bolt) using AutoGluon.⁴ The models process a sliding window of $L = 90$ daily closing prices to forecast a horizon of $h = 7$ days. For each window, we generate 30 Monte Carlo forecast paths to derive directional probabilities relative to the price thresholds defined in Section 4.

TimeGPT-1 [21] and TiRex [5] are applied in zero-shot mode. Their forecasts are mapped to directional signals using the same threshold logic.

Secondary Model Training The secondary model ($M2$) is a CNN-Transformer binary classifier trained on the meta labels to predict the reliability of $M1$'s directional forecasts. It processes the 90-day lookback window and follows the fixed structural template:

RevIN $\rightarrow$ 1D CNN $\rightarrow$ learnable positional embedding $\rightarrow$ Transformer encoder $\rightarrow$ pooling $\rightarrow$ 2-layer MLP head $\rightarrow$ reliability probability

Within this template, configuration-specific hyperparameters (layer widths, attention configurations, dropout rates, and activations) are selected per ($M1$, side, asset) combination based on validation performance. Across our experiments, the CNN uses channel widths in $\{64, 128\}$; the Transformer encoder has $\{2, 4\}$ layers with 2 or 4 attention heads, d_{model} in $\{64, 128\}$, and feed-forward width of $4 \times d_{\text{model}}$. The MLP classifier has a single hidden layer of size 128–256 (Linear $\rightarrow$ activation $\rightarrow$ dropout $\rightarrow$ Linear); pooling is mean-max, and activations are chosen from $\{\text{GELU}, \text{ReLU}\}$.

We train separate $M2$ instances for the Up-side and Down-side tasks, and validate our architectural contributions through an ablation study across three variants: (1) Vanilla CTTS [59], (2) CTTS + RevIN, and (3) CTTS + RevIN + Learnable Positional Encoding (see Appendix C and G).

Time-Series Cross-Validation To mitigate overfitting, which is a common challenge when applying deep learning to limited financial time series, we employ an expanding window cross-validation scheme [26]. The model is iteratively trained on an expanding dataset and evaluated on a subsequent fixed-size validation block. Careful attention must be paid to prevent information leakage by using appropriate data splits and time series aware cross-validation [33]. The final validation score is the average metric across all folds, ensuring the selected hyperparameters generalize well to different market regimes.

Reproducibility To facilitate reproduction, we report the specific foundation model checkpoints used: Chronos [chronos-t5-large], Chronos-Bolt [bolt-base], and TiRex, all accessed via their public releases on HuggingFace; TimeGPT-1 is accessed via the Nixtla hosted API. Fine-tuning hyperparameter ranges are reported in Table 1; the $M2$ architecture and configuration ranges are reported in Section 5.2 and the dataset splits are described in Section 5.1. Among the four primary models, three (Chronos, Chronos-Bolt, TiRex) are open-source and fully reproducible

from public checkpoints; TimeGPT-1 is the single proprietary component and depends on continued API availability.

5.3. Evaluation metrics

We assess performance using the following metrics, categorized into three types:

- **Classification Metrics:** We report standard Accuracy and Precision. Given the high cost of False Positives in trading, Precision is our primary classification metric.
- **Selective Classification Metrics:** We evaluate decision decoupling using Coverage and Risk. Lower Risk is better, though achieving it typically lowers Coverage.
- **Financial Metrics:** To validate economic viability, we simulate a simple trading strategy based on the model's decisions and report the Number of Trades, Cumulative Returns, Sharpe Ratio, Maximum Drawdown and Total Fees.

5.4. Baselines

To verify that the precision gains of $M2$ are not merely an artifact of conservative filtering, we compare against four alternative reliability gates. Each is applied as a second-stage filter on the same $M1$ predictions, uses the same chronological 70/15/15 meta-label split, and selects its threshold τ on the validation split under the same risk-coverage optimization rule used for $M2$. This isolates the architectural contribution of $M2$ from the choice of operating point.

Random Forest as a reliability model ($M1 + RF$).

A Random Forest classifier is trained on the meta labels defined in Definition 3.4 generated from the same $M1$ instance. The class-1 probability serves as the reliability score; $M1$'s signal is retained when this score exceeds τ .

Sequential predictive conformal inference ($M1 + SPCI$). A conformal prediction baseline derived from Xu and Xie [57]. SPCI is fitted on the same meta label split and produces prediction intervals (center, width) for the 7-step ahead realized price. We convert the interval into a directional margin score:

$$s_t^{\text{SPCI}} = \begin{cases} \frac{\text{center}_t - \text{close}_t}{\text{width}_t}, & \text{UP signal,} \\ \frac{\text{close}_t - \text{center}_t}{\text{width}_t}, & \text{DOWN signal.} \end{cases} \quad (11)$$

Positive values indicate that the conformal forecast center supports $M1$'s direction, with magnitude denoting the strength of support relative to conformal uncertainty. $M1$'s signal is retained when $s_t^{\text{SPCI}} \geq \tau$.

Chronos predictive interval gate ($M1 + PI$). A predictive variance baseline that uses $M1$'s native quantile forecasts and trains no second-stage model. For each Chronos signal we compute the relative interval width $w_t = \frac{(q_{0.9,t} - q_{0.1,t})}{C_t}$, where C_t is the current close, and set the reliability score $s_t^{\text{PI}} = -w_t$ so that narrower intervals score higher. $M1$'s signal is retained when $s_t^{\text{PI}} \geq \tau$.

Native-confidence gate ($M1 + \text{Conf}$). A confidence-thresholding baseline that reuses $M1$'s own directional confidence and trains no second-stage model. For each Chronos signal we take the empirical direction probability already used to assign the Up/Down label (Section 4.2), i.e., the fraction of the 30 Monte Carlo forecast paths that cross ρ in the predicted direction, as the reliability score $s_t^{\text{Conf}} \in [0, 1]$. $M1$'s signal is retained when $s_t^{\text{Conf}} \geq \tau$. Because this score is a monotonic summary of $M1$'s predictive distribution, any order-preserving post-hoc calibration (e.g., Platt scaling or isotonic regression) leaves the executed set, and hence Coverage and Risk, unchanged; $M1 + \text{Conf}$ therefore also stands in for the family of calibrated-probability thresholding gates.

⁴ <https://auto.gluon.ai/stable/tutorials/timeseries/index.html>.

Table 2

Comparison of alternative second-stage reliability estimators for BTC, ETH, SPY, QQQ, TSLA, and AMZN within the proposed two-stage architecture, using Chronos as the primary forecaster ($M1$). All variants share the same $M1$ predictions, the same chronological split, and solve the same risk–coverage optimization to select their operating threshold. ($M1$): standalone primary forecaster; ($M1 + M2$): the proposed reliability estimator; ($M1 + RF$): Random-Forest reliability estimator; ($M1 + SPCI$): conformal directional-margin reliability estimator; ($M1 + PI$): Chronos predictive-interval reliability estimator; ($M1 + Conf$): Chronos native-confidence-thresholding gate that reuses $M1$'s own directional confidence as the reliability score.

Asset	Model	UP				DOWN			
		Accuracy	Precision	Coverage	Risk	Accuracy	Precision	Coverage	Risk
BTC	$M1$	45%	42%	100%	58%	49%	47%	100%	53%
	$M1 + M2$	55%	60%	31%	40%	55%	86%	14%	14%
	$M1 + RF$	48%	42%	43%	57%	50%	40%	41%	60%
	$M1 + SPCI$	52%	48%	29%	52%	56%	73%	31%	27%
	$M1 + PI$	56%	59%	33%	41%	47%	42%	98%	58%
	$M1 + Conf$	43%	42%	100%	58%	50%	43%	61%	57%
ETH	$M1$	48%	50%	100%	50%	54%	50%	100%	50%
	$M1 + M2$	53%	64%	30%	36%	56%	58%	22%	42%
	$M1 + RF$	48%	50%	72%	50%	55%	51%	65%	49%
	$M1 + SPCI$	51%	59%	21%	41%	60%	57%	71%	43%
	$M1 + PI$	48%	46%	12%	54%	52%	48%	90%	52%
	$M1 + Conf$	52%	64%	21%	36%	48%	38%	44%	62%
SPY	$M1$	46%	59%	100%	41%	50%	32%	100%	68%
	$M1 + M2$	41%	63%	19%	37%	57%	35%	62%	65%
	$M1 + RF$	41%	69%	15%	31%	57%	31%	56%	69%
	$M1 + SPCI$	43%	62%	41%	38%	60%	34%	54%	66%
	$M1 + PI$	40%	75%	5%	25%	66%	31%	7%	69%
	$M1 + Conf$	39%	50%	2%	50%	68%	33%	7%	67%
QQQ	$M1$	47%	60%	100%	40%	44%	33%	100%	67%
	$M1 + M2$	41%	60%	34%	40%	61%	43%	52%	57%
	$M1 + RF$	43%	59%	61%	41%	53%	31%	57%	69%
	$M1 + SPCI$	43%	64%	41%	36%	55%	35%	55%	65%
	$M1 + PI$	44%	70%	34%	30%	45%	33%	99%	67%
	$M1 + Conf$	47%	60%	100%	40%	61%	29%	20%	71%
TSLA	$M1$	51%	53%	100%	47%	47%	42%	100%	58%
	$M1 + M2$	52%	82%	11%	18%	54%	50%	20%	50%
	$M1 + RF$	51%	53%	88%	47%	53%	47%	52%	53%
	$M1 + SPCI$	55%	79%	23%	21%	57%	54%	72%	46%
	$M1 + PI$	52%	81%	12%	19%	52%	22%	8%	78%
	$M1 + Conf$	50%	61%	17%	39%	56%	59%	19%	41%
AMZN	$M1$	50%	56%	100%	44%	47%	43%	100%	57%
	$M1 + M2$	47%	62%	25%	38%	58%	59%	14%	41%
	$M1 + RF$	45%	67%	2%	33%	60%	59%	27%	41%
	$M1 + SPCI$	55%	70%	63%	30%	60%	56%	51%	44%
	$M1 + PI$	44%	49%	49%	51%	57%	62%	3%	38%
	$M1 + Conf$	50%	56%	100%	44%	57%	67%	3%	33%

6. Experimental results

In this section, we present the performance of our approach. We then conduct an ablation study and backtesting to validate our findings. Finally, we provide a discussion of the results.

6.1. Selective forecasting performance

We begin by evaluating our **core hypothesis**: the two-stage learning architecture ($M1 + M2$) improves the precision and reduces the risk of forecasts compared to the standalone primary model ($M1$).

Main Results Table 2 confirms our hypothesis consistently across the studied assets. The combined model ($M1 + M2$), utilizing fine-tuned Chronos as the primary predictor and our improved CNN-Transformer as the secondary reliability estimator, achieves higher precision than $M1$ in almost all scenarios (the only exception being the QQQ up-side task, where precision remains unchanged at 60%).

Across all assets and all models, adding the decision layer increases precision on average by 12% on the Upside and 16% on the Downside (14% overall), thereby reducing risk by 12–16%. The gains are slightly larger on the Downside, suggesting that the reliability layer is especially effective at filtering short signals. As expected, this increase in precision and reduction in risk comes at the cost of coverage (execution rate). This trade-off is intentional and aligns with professional trading practices, where capital is deployed only on high-confidence signals. By

filtering out uncertain predictions, the system sacrifices trade frequency to maximize the reliability of executed trades.

Notably, there is an isolated case where this two-stage learning architecture does not outperform the baseline for the QQQ Up-Trend task (where precision remains unchanged). However, this deviation is specific to the Chronos $M1$ configuration. As detailed in the Appendix in Table D.3, where alternative foundation models (Chronos-Bolt, TimeGPT-1, TiRex) are employed as $M1$, the reliability model consistently improves Precision and reduces Risk across these assets.

Fig. 4 shows an example of selective execution on the Bitcoin dataset. The left plot shows the case where $M1$ makes a correct directional prediction (Up) and $M2$ correctly identifies it as reliable, triggering an execution. The right plot illustrates a case where $M1$ makes an incorrect prediction, but $M2$ detects the low reliability and successfully prevents execution (Abstention). The visualization of the other assets can be found in Fig. E.1 in the Appendix.

Beyond the standalone primary model, we compare $M2$ against four alternative second-stage reliability estimators that share the same $M1$ predictions and the same risk–coverage optimization: a Random-Forest gate ($M1 + RF$), a conformal directional-margin gate ($M1 + SPCI$), a Chronos predictive-interval gate ($M1 + PI$), and a Chronos native-confidence gate ($M1 + Conf$). Across assets and sides, $M2$ delivers the most consistent precision improvement and risk reduction. This comparison is not specific to Chronos: the two backbone-agnostic gates

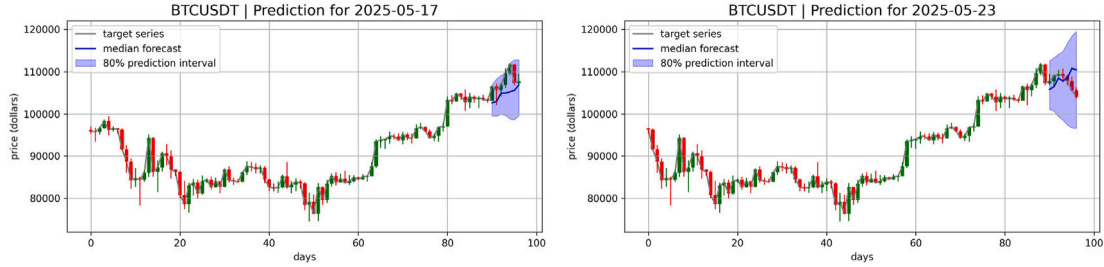


Fig. 4. Examples of selective execution on Bitcoin data. *Left:* A correct Up prediction validated by $M2$. *Right:* An incorrect Up prediction successfully filtered out by $M2$, demonstrating the system's ability to mitigate risk.

($M1 + RF$ and $M1 + SPCI$) are evaluated against $M2$ across all four primary backbones (Chronos, Chronos-Bolt, TimeGPT-1, TiRex) and all six assets in Appendix D (Table D.3), where $M2$ remains the most consistent reliability gate across backbones.

Risk–Coverage Optimization Analysis The Risk–Coverage curves for each asset are presented in Fig. G.3 in the Appendix G. As anticipated, higher coverage generally entails a trade-off with increased risk. These curves also serve as a sensitivity analysis over the operating point: rather than relying on a single selected threshold, they trace risk across the full range of coverage levels, so the reliability behavior can be assessed independently of any particular threshold choice.

The curves make the trade-off explicit: each decision threshold θ corresponds to a specific (Coverage, Risk) operating point, and the practitioner can position the system anywhere along the achievable frontier. Our constrained optimization (Section 4.4) selects θ by maximizing coverage subject to a risk cap $\alpha \leq 20\%$ and a minimum coverage floor $C_{\min} \geq 5\%$. The variation in curve shapes across assets reflects the heterogeneous reliability structure that $M2$ learns from each (asset, side) combination.

6.2. Ablation study

To validate the effectiveness of our architectural components, we compare the vanilla CTTS (CNN-Transformer) model against variants incorporating these modules. Table 3 reports the ablation results averaged across the six assets; per-asset breakdown results are reported in Appendix C, Table C.2 and the corresponding risk–coverage curves are reported in Appendix G.

For the Up-trend task, the addition of RevIN and Learnable Positional Encodings boosts Precision significantly, rising from 52.8% to 65.2%, while simultaneously reducing Risk from 47.2% to 34.8%. Although this stricter filtering leads to lower Coverage, the architecture effectively achieves the primary objective of risk mitigation. For the Down-trend task, the architectural improvements drive comparable gains, increasing Precision from 38.0% to 55.2% and reducing Risk from 62.0% to 44.8%, with a modest Accuracy improvement from 52.5% to 56.8%.

We additionally report the validation risk at a fixed 20% coverage (Risk@20% Cov.), a fixed-coverage comparison that isolates the architectural contribution from a more conservative operating threshold. At matched coverage, the proposed variant attains the lowest overall risk (33.3%, versus 37.1% and 41.3% for the simpler variants) and the lowest up-side risk (21.7%); on the down side the variants are comparable, consistent with the greater down-side variability discussed in Appendix G.

Overall, the results demonstrate that addressing non-stationarity (via RevIN) and translation invariance (via Learnable Positional Encodings) are critical for improving the confidence and reliability of the secondary model.

6.3. Backtesting results

To assess the economic viability of our approach, we implement a long–short trading strategy driven by the consensus signal of the combined $M1 + M2$ system. The trading strategy logic is the following:

Table 3

Mean ablation performance across the six assets with Chronos as $M1$. We compare the baseline CTTS against variants incorporating reversible instance normalization (RevIN) and learnable positional encodings. Risk@20% Cov. reports validation risk at fixed 20% coverage from the risk–coverage curves, included to separate architecture effects from threshold conservativeness.

Side	Model	Accuracy	Precision	Coverage	Risk	Risk@20% Cov.
UP	Original CTTS	46.3%	52.8%	42.5%	47.2%	30.2%
	+ RevIN	46.3%	51.3%	55.2%	48.7%	28.8%
	+ RevIN & Learn. PE (ours)	48.2%	65.2%	25.0%	34.8%	21.7%
DOWN	Original CTTS	52.5%	38.0%	49.0%	62.0%	44%
	+ RevIN	54.0%	42.2%	60.5%	57.8%	53.8%
	+ RevIN & Learn. PE (ours)	56.8%	55.2%	30.7%	44.8%	45%
Overall	Original CTTS	49.4%	45.4%	45.8%	54.6%	37.1%
	+ RevIN	50.2%	46.8%	57.8%	53.2%	41.3%
	+ RevIN & Learn. PE (ours)	52.5%	60.2%	27.8%	39.8%	33.3%

A position is opened only if both $M1$ and $M2$ agree on the direction. Position size is scaled by the secondary model's reliability score following the bet-sizing rule from De Prado [14]. Given a directional signal of side $s \in \{+1, -1\}$ (long, short) and reliability score $\pi_t \in (0, 1)$, we compute the z-statistic

$$\zeta_t = \frac{\pi_t - 0.5}{\sqrt{\pi_t(1 - \pi_t)}} \quad (12)$$

and map it to a signed bet size $m_t = s \cdot (2\Phi(\zeta_t) - 1) \in [-1, 1]$, where Φ is the standard normal CDF. The notional trade value is $V_t = w_{\max} \cdot \text{Cash}_t \cdot |m_t|$, with $w_{\max} = 0.02$ and Cash_t denoting available cash at time t . Each new position is therefore capped at 2% of available cash, with higher reliability producing larger exposure. We do not impose an explicit cap on the number of concurrent positions; the binding constraint is available cash.

Trades are held for a maximum of 7 days, with fixed take-profit rules (10% for Crypto, 5% for Equities) and early-exit conditions based on price reversal signals. Due to platform limitations on QuantConnect,⁵ we run separate backtests for the Crypto (BTC, ETH) and the Equities (SPY, QQQ, TSLA, AMZN). Table 4 summarizes the performance metrics for both asset types.

For crypto assets, the selective $M1 + M2$ strategy delivers a + 20.3% cumulative return with a Sharpe ratio of 2.38 and a remarkably low Maximum Drawdown of 5.8%. This indicates that the reliability model effectively identifies high-quality signals in the volatile crypto market, successfully concentrating risk where the probability of profit is highest.

For equities, the strategy achieves a cumulative return of 39.0%. However, it exhibits a lower Sharpe ratio (0.94) and a higher maximum

⁵ <https://www.quantconnect.com/>.

Table 4

Performance of the selective $M1 + M2$ long-short strategy applied to the crypto and equity classes.

	Crypto	Equities
Number of trades	123	605
Cumulative Returns	20.32%	39.03%
Sharpe ratio	2.38	0.94
Max drawdown	5.80%	9.00%
Turnover	18.51%	25.43%
Total fees (of initial equity)	3.26%	0.60%

drawdown (9.0%) compared to crypto. This reflects the different market dynamics of equities, where trends may be noisier or harder to separate from general market beta.

Fig. 5 illustrates the cumulative return curves, comparing the selective $M1 + M2$ strategy against the $M1$ -only baseline (100% execution) and the Buy-and-Hold benchmark. The curves demonstrate that selective execution reduces and limits drawdowns, helping to protect capital during market downturns.

Our backtest applies each brokerage's actual fee schedule (Binance for crypto; Interactive Brokers for equities), an upper bound on retail execution costs. We do not separately model bid-ask spreads and slippage, but their effect is bounded: the traded assets are highly liquid (major cryptocurrencies and large-cap equities), with quoted spreads on the order of the per-trade fees already deducted. As a conservative check, doubling the realized per-trade frictions—adding a spread-and-slippage cost equal in magnitude to the fees already absorbed—lowers cumulative returns only modestly, from 20.3% to about 17.1% for crypto and from 39.0% to about 38.4% for equities, leaving both portfolios clearly profitable. Market impact is negligible at the tested \$10k scale, where order sizes are a tiny fraction of these assets' average daily volume. Extended financial metrics (Sortino ratio, Probabilistic Sharpe, expectancy, profit factor, Value-at-Risk) and monthly rolling returns spanning the full test windows, 20 months for

equities (including 7 negative-return months) and 5 months for crypto, are reported in Appendix F.

6.4. Discussion

In this subsection, we present some observations from our experiments and discuss their practical implications.

Fine-Tuned vs. Zero-Shot Foundation Models Our experiments cover both fine-tuned models (Chronos, Chronos-Bolt) and zero-shot models (TimeGPT-1, TiRex). The two-stage architecture improves selective forecasting reliability for both paradigms: even when the underlying $M1$ signal is weak, $M2$ learns to recognize unreliable predictions and filters out many of the largest errors. This indicates that the architecture is model-agnostic and can be applied effectively regardless of the primary forecasting paradigm. Within both paradigms, however, performance varies considerably across assets. The detailed per-asset results in Table D.3 (Appendix D) reveal no systematic ranking; for some assets the zero-shot models outperform the fine-tuned ones, and for others the reverse holds. We therefore do not advocate for either paradigm as universally superior. The choice of $M1$ should depend on operational considerations such as the availability of fine-tuning data, compute budget, and the statistical characteristics of the target asset.

Gain Analysis Across Different Primary Models Beyond comparing paradigms, we examine how much $M2$ improves each individual $M1$. Table 5 summarizes $M2$'s per-model gains, averaged over the six assets and decomposed by direction. $M1 + M2$ improves precision over standalone $M1$ in 46 of the 48 (asset, $M1$ model, side) combinations, with two exceptions: a tie on Chronos for QQQ-UP and a small decrease on TimeGPT-1 for SPY-UP. Precision gains range from +10.8 to +18.1 percentage points, with Chronos-Bolt benefiting most across both directions and Down-side gains consistently exceeding UP-side gains for every $M1$ model.

The two-stage framework therefore yields consistent improvements across primary forecasters (its value is not tied to a specific $M1$ model) while the magnitude of the gain is model-dependent, suggesting that

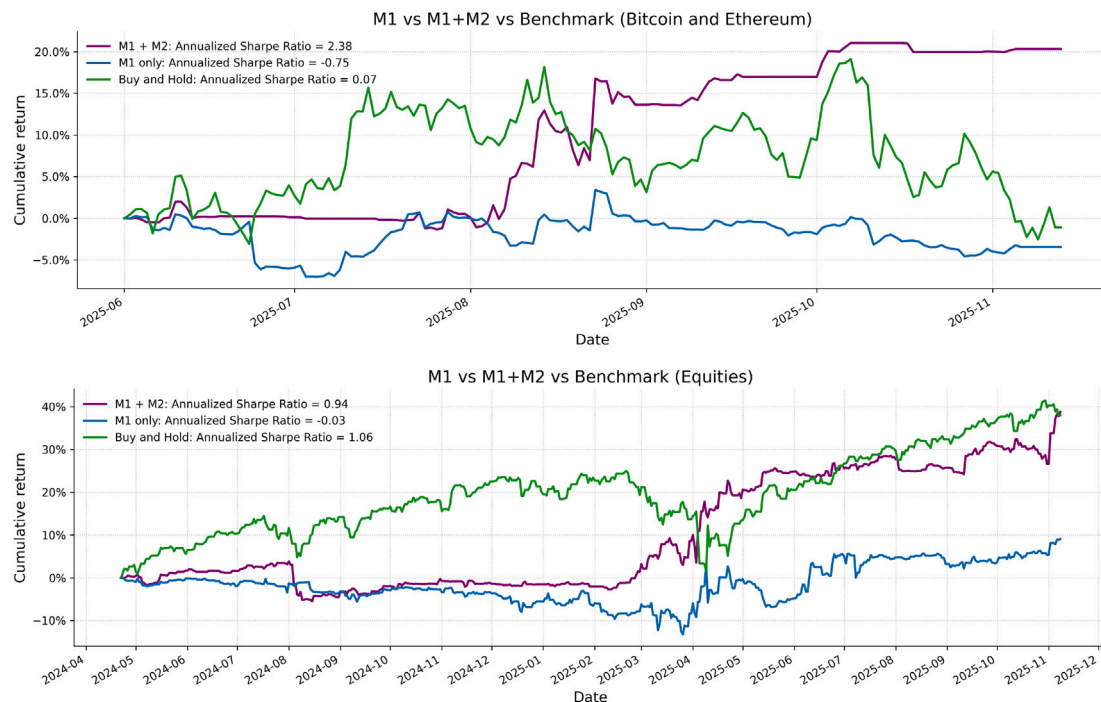


Fig. 5. Backtesting results: cumulative return curves for the meta-labeled trading strategy ($M1 + M2$) compared against the standalone primary model ($M1$, 100% execution) and a Buy-and-Hold benchmark. *Top*: cryptocurrency portfolio (BTC, ETH). *Bottom*: equity portfolio (SPY, QQQ, TSLA, AMZN). Legends include the corresponding annualized Sharpe ratios for each strategy.

Table 5

Precision-gain summary for each $M1$ model with paired moving-block bootstrap uncertainty (5000 resamples, block length 7, matching the 7-day prediction horizon). Values are averaged across the six assets. Δ Prec. is Precision ($M1 + M2$) - Precision ($M1$) in percentage points; Cov. is the mean selective coverage of $M1 + M2$ as a percentage. The Overall column reports the mean Δ Precision across both sides with a 95% bootstrap CI. It is constructed by averaging Δ Precision across UP and DOWN within each resample and taking the 2.5th and 97.5th percentiles. P-values are derived from the same bootstrap distribution: $*p < 0.05$, $**p < 0.01$; cells without a marker have $p \geq 0.05$.

$M1$ model	UP		DOWN		Overall
	Δ Prec.	Cov.	Δ Prec.	Cov.	Δ Prec. [95% CI]
Chronos	+11.8*	24.3	+13.5*	29.4	+12.6 [+4.9, +18.6]
Chronos-Bolt	+14.4**	19.6	+18.1**	34.1	+16.3 [+8.7, +24.8]
TimeGPT-1	+11.8**	37.7	+16.3*	24.1	+14.1 [+6.6, +21.0]
TiRex	+10.8	24.1	+14.6**	38.2	+12.7 [+4.6, +19.3]

the learnability of reliability signals varies with the choice of $M1$. These gains are achieved under selective execution, with mean $M1 + M2$ coverage between 20% and 38%, consistent with the intended risk-coverage trade-off. Paired moving-block bootstrap CIs (Table 5) confirm that 7 of the 8 (model, side) precision gains are statistically significant at $p < 0.05$; the sole non-significant cell is TiRex-UP (+10.8%, $p \geq 0.05$), where the point estimate remains positive but the CI marginally includes zero. All four $M1$ models show an Overall Δ Precision CI strictly above zero.

Economic Rationale for Low Coverage A recurring theme in our results is the trade-off between Risk and Coverage. Adding the secondary model significantly improved the precision but at the cost of lower coverage. From a practical trading perspective, low coverage is not necessarily a drawback, particularly in a swing-trading context. A swing trader does not need to trade the market every day; rather, the goal is to deploy capital only when the probability of success is maximized.

In a multi-asset setting, low coverage per asset translates into a meaningful overall trade flow once signals are aggregated across a diverse set of assets. In this context, the Risk-Coverage pair becomes a natural input for portfolio construction. We can allocate more capital to high-probability predictions identified by $M2$, while scaling down exposure to uncertain regimes. Our framework makes this trade-off explicit: the strategic benefit of acting less often, but with inherently lower risk, is particularly valuable for capital preservation and risk-managed portfolios. What the reliability layer provides is risk-adjusted quality rather than raw market exposure: more reliable executed trades and lower drawdown (5.8% crypto, 9.0% equities), which matters most in volatile or uncertain regimes. In a sustained uptrend like the equity test window, trading selectively naturally gives up some upside compared to staying fully invested—the trade-off is visible against the buy-and-hold line in Fig. 5.

Interpretability of the Reliability Estimator As an exploratory analysis, we inspect $M2$'s self-attention to see which parts of the input window are emphasized when accepting versus rejecting an $M1$ forecast. Aggregated over $M2$'s decisions on the test set, accept decisions weight the most recent input tokens more heavily than reject decisions, which spread attention more evenly. In accordance with common practice in attention-based diagnostics [29], we understand these visualizations as providing qualitative insight into the temporal regions considered important by the reliability estimator, rather than causal explanations of $M2$'s behavior; full details and visualizations are given in Appendix H.

Hyperparameter Sensitivity We examined how $M2$'s training hyperparameters affect validation performance using the Optuna trial data from our study with Chronos as $M1$. Learning rate and weight decay emerge as the most influential hyperparameters (Spearman correlation $\approx +0.50$, $p < 0.05$ for both), while batch size shows no statistically significant effect over the tested range. The best-performing configuration uses a learning rate $\approx 5 \times 10^{-4}$ and weight decay $\approx 10^{-4}$.

7. Conclusions

In this paper, we introduce a two-stage learning architecture for selective short-term financial time series trend forecasting. By decoupling the forecasting task (primary model) from the reliability estimation task (secondary model), we establish a risk-controlled prediction framework capable of handling the noise, low signal-to-noise ratio and non-stationarity inherent in financial markets.

Our empirical evaluation studies state-of-the-art foundation models as primary models, including Chronos, Chronos-Bolt, TimeGPT-1 and TiRex, across a diverse set of equities and cryptocurrencies. As a secondary model, we enhance a CNN-Transformer with Reversible Instance Normalization (RevIN) and Learnable Positional Encodings. The results demonstrate that our proposed two-stage learning architecture significantly boosts precision and reduces risk compared to standalone forecasters. That said, even though bootstrap shows the precision gains are not sampling artifacts, it does not replace a rolling-window or regime-specific evaluation, and financial results are sensitive to the chosen period—the main limitation of this study. Since our windows include corrections and several down months but no sustained bear or stress regime, reliability under crisis conditions remains suggestive; we will therefore evaluate the framework across rolling windows and out-of-distribution stress regimes (e.g., the 2020 COVID crash or the 2022 inflation shock) to assess robustness across market regimes. In future work, we want to identify the fundamental drivers (features) behind short-term directional movements, for each direction. This will allow us to leverage our distinction between Up and Down models. An interesting aspect is to incorporate causally related exogenous variables [27]. Further, instead of using fixed thresholds, we plan to adapt the Up and Down decision thresholds to the asset's current volatility. In more volatile regimes, the model would require larger moves to call an Up or Down signal, while in calmer markets smaller moves would be sufficient. Finally, developing finite-sample guarantees for risk-constrained selective execution under non-stationarity is a meaningful direction for future theoretical work. We hope that our approach supports the development of automated trading systems that prioritize reliability over raw prediction coverage, thereby reducing overtrading and improving decision quality under transaction costs and market uncertainty.

CRedit authorship contribution statement

Albi Isufaj: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Pablo Mollá:** Writing – review & editing, Validation, Software, Methodology, Investigation, Formal analysis. **Jing Li:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Yutaka Matsuo:** Supervision, Funding acquisition, Conceptualization. **Helmut Prendinger:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Two-stage learning architecture algorithm

Algorithm 1: Two-stage learning pipeline for financial time series trend forecasting. *Stage 1 (training & threshold selection):* Train $M1$, generate side-specific meta-label datasets, train $M2^{(s)}$, and choose $\theta^{(s)}$ on validation by risk-constrained coverage maximization. *Stage 2 (test-time selective execution):* Execute a directional signal only when its side-specific reliability score passes the frozen threshold.

Inputs: Labeled splits D_{train} and D_{meta} (with $D_{meta} \cap D_{train} = \emptyset$), where D_{meta} is the held-out 40% segment on which the frozen $M1$ generates predictions, split chronologically 70/15/15 into $D_{meta,train}$, $D_{meta,val}$, $D_{meta,test}$ ($D_{meta,train}$ and $D_{meta,val}$ supply the meta-labels for training and selecting $M2$; $D_{meta,test}$ is the held-out test set). Lookback L , horizon h , movement threshold ρ , risk cap α , minimum coverage C_{min} .

Outputs: Frozen $M1$, side-specific reliability models $M2^{(Up)}, M2^{(Down)}$, thresholds $\theta^{(Up)}, \theta^{(Down)}$, final selective decisions $\hat{y}_t^{final}$ on $D_{meta,test}$.

1 Phase 1: Meta-label generation, side-specific reliability training, and threshold selection

```

3 Train primary forecaster  $M1$  on  $D_{train}$  and freeze  $M1$ ;
5 Chronologically split  $D_{meta}$  into  $D_{meta,train}$ ,  $D_{meta,val}$ ,  $D_{meta,test}$  using a 70/15/15 ratio;
6 for  $s \in \{Up, Down\}$  do
7   Initialize side-specific meta sets  $\mathcal{D}_{meta,train}^{(s)}, \mathcal{D}_{meta,val}^{(s)}$ ;
8   foreach  $t \in D_{meta,train}$  do
9     Compute primary prediction  $\hat{y}_t$  from  $M1$ ;
10    if  $\hat{y}_t = s$  then
11      Compute meta-label  $z_t = \mathbb{1}(\hat{y}_t = y_t)$ ;
12      Add  $(X_t, z_t)$  to  $\mathcal{D}_{meta,train}^{(s)}$ ;
13    end
14  end
15  Train  $M2^{(s)}$  on  $\mathcal{D}_{meta,train}^{(s)}$  to estimate reliability score  $\pi_t^{(s)} = P(z_t = 1 \mid X_t, s)$ ;
16  foreach  $t \in D_{meta,val}$  do
17    Compute primary prediction  $\hat{y}_t$  from  $M1$ ;
18    if  $\hat{y}_t = s$  then
19      Compute reliability score  $\pi_t^{(s)} = M2^{(s)}(X_t)$  and store  $(\pi_t^{(s)}, y_t, \hat{y}_t)$  in  $\mathcal{D}_{meta,val}^{(s)}$ ;
20    end
21  end
22  Define candidate thresholds  $\Theta^{(s)}$  from unique values of  $\pi_t^{(s)}$  on  $\mathcal{D}_{meta,val}^{(s)}$ ;
23  Define feasible set  $\Theta_{feas}^{(s)} = \{\theta \in \Theta^{(s)} : \text{Risk}_{val}^{(s)}(\theta) \leq \alpha, \text{Coverage}_{val}^{(s)}(\theta) \geq C_{min}\}$ ;
24  if  $\Theta_{feas}^{(s)} = \emptyset$  then
25    Prune this trial;
26  else
27    Select  $\theta^{(s)} = \arg \max_{\theta \in \Theta_{feas}^{(s)}} \text{Coverage}_{val}^{(s)}(\theta)$ ;
28  end
29 end

```

30 Phase 2: Test-time selective execution

```

31 foreach  $t \in D_{meta,test}$  do
32   Compute primary prediction  $\hat{y}_t \in \{Up, Down, Same\}$  from  $M1$ ;
33   if  $\hat{y}_t = Same$  then
34      $\hat{y}_t^{final} \leftarrow \text{Abstain}$ ;
35   else
36     Set side  $s \leftarrow \hat{y}_t$ ;
37     Compute reliability score  $\pi_t^{(s)} = M2^{(s)}(X_t)$ ;
38     if  $\pi_t^{(s)} \geq \theta^{(s)}$  then
39        $\hat{y}_t^{final} \leftarrow s$ ;
40     else
41        $\hat{y}_t^{final} \leftarrow \text{Abstain}$ ;
42     end
43   end
44 end
45 return  $\{\hat{y}_t^{final}\}_{t \in D_{meta,test}}$ ;

```

Appendix B. Dataset date ranges

Table B.1

Date ranges (all end on Nov 11, 2025) for the full datasets used in this study. Assets are ordered by market capitalization (descending).

	CRYPTO			EQUITIES		
	Start date	Trading days	Asset	Start date	Trading days	
1	BTCUSDT	2018-02-01	2842	SPY	2000-01-03	6506
2	ETHUSDT	2018-02-01	2842	QQQ	2000-01-03	6506
3	XRPUSDT	2018-05-04	2750	NVDA	2000-01-03	6506
4	BNBUSDT	2018-02-01	2842	AAPL	2000-01-03	6506
5	SOLUSDT	2020-08-11	1920	GOOG	2004-08-19	5344
6	TRXUSDT	2018-06-11	2712	MSFT	2000-01-03	6506
7	DOGEUSDT	2019-07-05	2323	AMZN	2000-01-03	6506
8	ADAUSDT	2018-04-17	2767	AVGO	2009-08-06	4094
9	BCHUSDT	2019-11-28	2177	META	2012-05-18	3392
10	LINKUSDT	2019-01-16	2493	TSLA	2010-06-29	3869
11	XTMUSDT	2018-05-31	2723	LLY	2000-01-03	6506
12	LTCUSDT	2018-02-01	2842	WMT	2000-01-03	6506
13	SUIUSDT	2023-05-03	925	JPM	2000-01-03	6506
14	AVAXUSDT	2020-09-22	1878	V	2008-03-19	4443
15	HBARUSDT	2019-09-29	2237	XOM	2000-01-03	6506
16	SHIBUSDT	2021-05-10	1648	MA	2006-05-25	4899
17	TONUSDT	2024-08-08	462	JNJ	2000-01-03	6506
18	DOTUSDT	2020-08-18	1913	NFLX	2002-05-23	5908
19	AAVEUSDT	2020-10-15	1855	ABBV	2013-01-02	3237
20	NEARUSDT	2020-10-14	1856	COST	2000-01-03	6506

Appendix C. Extended ablation study

Table C.2

Extended ablation study covering the six assets (BTC, ETH, SPY, QQQ, TSLA, AMZN). We compare the baseline CTTS against variants incorporating reversible instance normalization (RevIN) and our proposed architecture (CTTS + RevIN + Learnable Positional Encoding).

Asset	Model	UP				DOWN			
		Accuracy	Precision	Coverage	Risk	Accuracy	Precision	Coverage	Risk
BTC	Original CTTS	52%	44%	14%	56%	50%	44%	51%	56%
	+ RevIN	51%	47%	55%	53%	51%	44%	33%	56%
	+ RevIN & Learn. PE (ours)	55%	60%	31%	40%	55%	86%	14%	14%
ETH	Original CTTS	50%	54%	26%	46%	55%	51%	83%	49%
	+ RevIN	48%	50%	10%	50%	<u>56%</u>	53%	90%	47%
	+ RevIN & Learn. PE (ours)	53%	64%	30%	36%	<u>56%</u>	58%	22%	42%
S&P 500	Original CTTS	42%	56%	54%	44%	53%	26%	59%	74%
	+ RevIN	38%	49%	38%	51%	54%	27%	58%	73%
	+ RevIN & Learn. PE (ours)	41%	63%	19%	37%	57%	35%	62%	65%
QQQ	Original CTTS	38%	49%	68%	51%	53%	28%	47%	72%
	+ RevIN	45%	58%	90%	42%	<u>61%</u>	38%	33%	62%
	+ RevIN & Learn. PE (ours)	41%	60%	34%	40%	<u>61%</u>	43%	52%	57%
TSLA	Original CTTS	50%	61%	17%	39%	49%	35%	32%	65%
	+ RevIN	51%	53%	68%	47%	50%	45%	80%	55%
	+ RevIN & Learn. PE (ours)	52%	82%	11%	18%	54%	50%	20%	50%
AMZN	Original CTTS	46%	53%	76%	47%	55%	44%	22%	56%
	+ RevIN	45%	51%	70%	49%	52%	46%	69%	54%
	+ RevIN & Learn. PE (ours)	47%	62%	25%	38%	58%	59%	14%	41%

Appendix D. Extended experimental results

Table D.3

Detailed performance comparison across the full assets (BTC, ETH, SPY, QQQ, TSLA, AMZN) covering four primary foundation models. For each asset, we compare the standalone primary forecaster ($M1$), the proposed two-stage learning system ($M1 + M2$), the random forest reliability gate ($M1 + RF$), and the conformal directional-margin gate ($M1 + SPCI$).

Asset	Model	UP				DOWN			
		Accuracy	Precision	Coverage	Risk	Accuracy	Precision	Coverage	Risk
BTC	Chronos	45%	42%	100%	58%	49%	47%	100%	53%
	Chronos + $M2$	55%	60%	31%	40%	55%	86%	14%	14%
	Chronos + RF	48%	42%	43%	57%	50%	40%	41%	60%
	Chronos + SPCI	52%	48%	29%	52%	56%	73%	31%	27%
	Chronos-Bolt	52%	50%	100%	50%	50%	48%	100%	52%
	Chronos-Bolt + $M2$	53%	57%	27%	40%	56%	67%	18%	33%
	Chronos-Bolt + RF	50%	33%	35%	67%	44%	37%	49%	63%
	Chronos-Bolt + SPCI	52%	50%	100%	50%	50%	48%	99%	52%
	TimeGPT-1	56%	51%	100%	49%	58%	54%	100%	46%
	TimeGPT-1 + $M2$	62%	72%	26%	28%	58%	75%	23%	25%
	TimeGPT-1 + RF	56%	52%	60%	48%	61%	68%	63%	32%
	TimeGPT-1 + SPCI	56%	51%	100%	49%	58%	54%	100%	46%
	TiRex	36%	37%	100%	63%	39%	29%	100%	71%
	TiRex + $M2$	55%	78%	9%	22%	50%	44%	33%	56%
ETH	Chronos	48%	50%	100%	50%	54%	50%	100%	50%
	Chronos + $M2$	53%	64%	30%	36%	56%	58%	22%	42%
	Chronos + RF	48%	50%	72%	50%	55%	51%	65%	49%
	Chronos + SPCI	51%	59%	21%	41%	60%	57%	71%	43%
	Chronos-Bolt	45%	47%	100%	53%	46%	41%	100%	59%
	Chronos-Bolt + $M2$	50%	56%	24%	43%	54%	55%	47%	45%
	Chronos-Bolt + RF	54%	57%	62%	43%	53%	52%	46%	48%
	Chronos-Bolt + SPCI	52%	62%	25%	38%	47%	42%	97%	58%
	TimeGPT-1	42%	45%	100%	55%	48%	40%	100%	60%
	TimeGPT-1 + $M2$	52%	65%	24%	35%	55%	59%	24%	41%
	TimeGPT-1 + RF	46%	48%	80%	52%	49%	35%	50%	65%
	TimeGPT-1 + SPCI	47%	43%	22%	57%	52%	44%	62%	56%
	TiRex	47%	49%	100%	51%	45%	41%	100%	59%
	TiRex + $M2$	50%	59%	19%	40%	56%	60%	35%	40%
SPY	Chronos	46%	59%	100%	41%	50%	32%	100%	68%
	Chronos + $M2$	41%	63%	19%	37%	57%	35%	62%	65%
	Chronos + RF	41%	69%	15%	31%	57%	31%	56%	69%
	Chronos + SPCI	43%	62%	41%	38%	68%	34%	54%	66%
	Chronos-Bolt	51%	64%	100%	36%	61%	38%	100%	62%
	Chronos-Bolt + $M2$	41%	85%	11%	15%	64%	43%	71%	57%
	Chronos-Bolt + RF	43%	58%	73%	42%	61%	24%	45%	76%
	Chronos-Bolt + SPCI	51%	64%	99%	36%	67%	75%	4%	25%
	TimeGPT-1	61%	66%	100%	34%	65%	45%	100%	55%
	TimeGPT-1 + $M2$	49%	65%	52%	35%	68%	60%	30%	40%
	TimeGPT-1 + RF	59%	66%	88%	34%	64%	35%	49%	65%
	TimeGPT-1 + SPCI	51%	67%	55%	33%	67%	60%	7%	40%
	TiRex	62%	66%	100%	34%	67%	54%	100%	46%
	TiRex + $M2$	50%	71%	40%	28%	67%	56%	27%	44%
	TiRex + RF	40%	76%	7%	24%	69%	85%	32%	15%
	TiRex + SPCI	59%	66%	94%	34%	65%	25%	20%	75%

(continued on next page)

Table D.3 (continued)

Asset	Model	UP				DOWN			
		Accuracy	Precision	Coverage	Risk	Accuracy	Precision	Coverage	Risk
QQQ	Chronos	47%	60%	100%	40%	44%	33%	100%	67%
	Chronos + $M2$	41%	60%	34%	40%	61%	43%	52%	57%
	Chronos + RF	43%	59%	61%	41%	53%	31%	57%	69%
	Chronos + SPCI	43%	64%	41%	36%	55%	35%	55%	65%
	Chronos-Bolt	60%	64%	100%	36%	63%	29%	100%	71%
	Chronos-Bolt + $M2$	43%	69%	21%	31%	65%	50%	43%	50%
	Chronos-Bolt + RF	42%	70%	13%	30%	63%	29%	100%	71%
	Chronos-Bolt + SPCI	57%	64%	88%	36%	63%	29%	100%	71%
	TimeGPT-1	56%	65%	100%	35%	59%	40%	100%	60%
	TimeGPT-1 + $M2$	45%	68%	35%	31%	66%	67%	17%	33%
	TimeGPT-1 + RF	42%	63%	33%	37%	62%	43%	75%	57%
	TimeGPT-1 + SPCI	43%	76%	18%	24%	65%	54%	12%	46%
	TiRex	62%	67%	100%	33%	66%	54%	100%	46%
	TiRex + $M2$	44%	70%	25%	30%	68%	75%	49%	25%
	TiRex + RF	52%	64%	74%	36%	66%	69%	22%	31%
	TiRex + SPCI	51%	79%	33%	21%	65%	0%	0%	100%
TSLA	Chronos	51%	53%	100%	47%	47%	42%	100%	58%
	Chronos + $M2$	52%	82%	11%	18%	54%	50%	20%	50%
	Chronos + RF	51%	53%	88%	47%	53%	47%	52%	53%
	Chronos + SPCI	55%	79%	23%	21%	57%	54%	72%	46%
	Chronos-Bolt	49%	50%	100%	50%	54%	49%	100%	51%
	Chronos-Bolt + $M2$	54%	93%	9%	7%	56%	80%	14%	20%
	Chronos-Bolt + RF	51%	75%	7%	25%	55%	53%	41%	47%
	Chronos-Bolt + SPCI	54%	73%	16%	27%	56%	57%	81%	43%
	TimeGPT-1	48%	47%	100%	53%	49%	42%	100%	58%
	TimeGPT-1 + $M2$	54%	68%	24%	31%	57%	55%	37%	45%
	TimeGPT-1 + RF	44%	39%	67%	61%	57%	70%	12%	30%
	TimeGPT-1 + SPCI	48%	47%	100%	53%	55%	49%	46%	51%
	TiRex	52%	53%	100%	47%	55%	52%	100%	48%
	TiRex + $M2$	50%	56%	27%	44%	58%	65%	46%	35%
	TiRex + RF	53%	56%	60%	44%	56%	55%	83%	45%
	TiRex + SPCI	54%	70%	23%	30%	55%	52%	82%	48%
AMZN	Chronos	50%	56%	100%	44%	47%	43%	100%	57%
	Chronos + $M2$	47%	62%	25%	38%	58%	59%	14%	41%
	Chronos + RF	45%	67%	2%	33%	60%	59%	27%	41%
	Chronos + SPCI	55%	70%	63%	30%	60%	56%	51%	44%
	Chronos-Bolt	54%	57%	100%	43%	54%	42%	100%	58%
	Chronos-Bolt + $M2$	48%	60%	28%	40%	57%	62%	12%	38%
	Chronos-Bolt + RF	55%	59%	85%	41%	54%	41%	67%	59%
	Chronos-Bolt + SPCI	47%	71%	9%	29%	56%	48%	89%	52%
	TimeGPT-1	56%	57%	100%	43%	56%	45%	100%	55%
	TimeGPT-1 + $M2$	60%	66%	68%	34%	59%	50%	16%	50%
	TimeGPT-1 + RF	49%	57%	11%	43%	59%	49%	77%	51%
	TimeGPT-1 + SPCI	56%	72%	31%	28%	59%	50%	2%	50%
	TiRex	57%	60%	100%	40%	55%	39%	100%	61%
	TiRex + $M2$	52%	64%	27%	35%	60%	60%	38%	40%
	TiRex + RF	56%	59%	91%	41%	62%	73%	31%	27%
	TiRex + SPCI	53%	79%	18%	21%	55%	39%	100%	61%

Appendix E. Selective execution examples



Fig. E.1. Example of selective execution on different assets. For each asset, *Left*: a correct UP prediction validated by *M2*. *Middle*: a correct DOWN prediction validated by *M2*. *Right*: an incorrect prediction filtered out by *M2*, demonstrating the system's ability to mitigate risk.

Appendix F. Extended backtest analysis

This appendix reports extended financial metrics for the selective $M1 + M2$ strategy, together with monthly rolling returns for the crypto and equity portfolios, in Tables F.4–F.6.

Table F.4

Extended financial metrics for the selective $M1 + M2$ strategy.

Metric	Crypto	Equities
Test period	Jun–Nov 2025 (5 months)	Apr 2024–Nov 2025 (20 months)
Number of trades	123	605
Net profit	20.32%	39.03%
Compounding annual return	50.61%	23.74%
Sharpe ratio	2.38	0.94
Sortino ratio	4.66	1.14
Probabilistic Sharpe ratio	83.5%	68.2%
Expectancy	0.59	0.26
Profit factor	2.38	1.52
Win rate	47.5%	50.2%
Profit/loss ratio	2.34	1.51
Max drawdown	5.80%	9.00%
Drawdown recovery (days)	57	214
Value-at-Risk (95%)	−1.6%	−1.5%
Value-at-Risk (99%)	−2.4%	−2.2%
Portfolio turnover	18.51%	25.43%
Total fees (% of initial equity)	3.26%	0.60%
Fees as % of gross profit	12.14%	1.56%

Table F.5

Monthly rolling returns for the crypto portfolio.

Month	Return	Sharpe	Max DD
Jun 2025	+0.13%	−0.31	1.9%
Jul 2025	+0.41%	0.04	2.0%
Aug 2025	+13.53%	11.63	5.8%
Sep 2025	+2.93%	6.83	0.3%
Oct 2025	+2.63%	1.66	0.9%
Nov 2025 (partial)	+0.26%	3.47	0.0%

Table F.6

Monthly rolling returns for the equity portfolio. It recorded 7 negative-return months out of 20 across the test window. The strategy nonetheless ended + 39.03%. This indicates that gains are not concentrated in a single favorable regime.

Month	Return	Sharpe	Max DD	Month	Return	Sharpe	Max DD
Apr 2024	+0.16%	0.19	0.3%	Feb 2025	+5.41%	8.11	0.6%
May 2024	+1.21%	0.57	2.4%	Mar 2025	+6.32%	3.66	5.7%
Jun 2024	−0.39%	−4.37	0.8%	Apr 2025	+7.86%	4.57	5.8%
Jul 2024	+1.54%	1.80	0.7%	May 2025	+3.52%	3.93	0.9%
Aug 2024	−6.93%	−2.76	9.0%	Jun 2025	+1.11%	0.57	1.4%
Sep 2024	+1.43%	1.46	1.6%	Jul 2025	+2.19%	2.38	1.1%
Oct 2024	+1.66%	2.37	0.8%	Aug 2025	−1.53%	−0.65	2.2%
Nov 2024	−0.61%	−2.02	1.4%	Sep 2025	+4.18%	3.40	1.2%
Dec 2024	−0.18%	−2.12	0.8%	Oct 2025	−3.24%	−3.10	4.3%
Jan 2025	−0.53%	−3.05	1.1%	Nov 2025 (partial)	+3.92%	24.37	0.4%

Appendix G. Risk–coverage curves

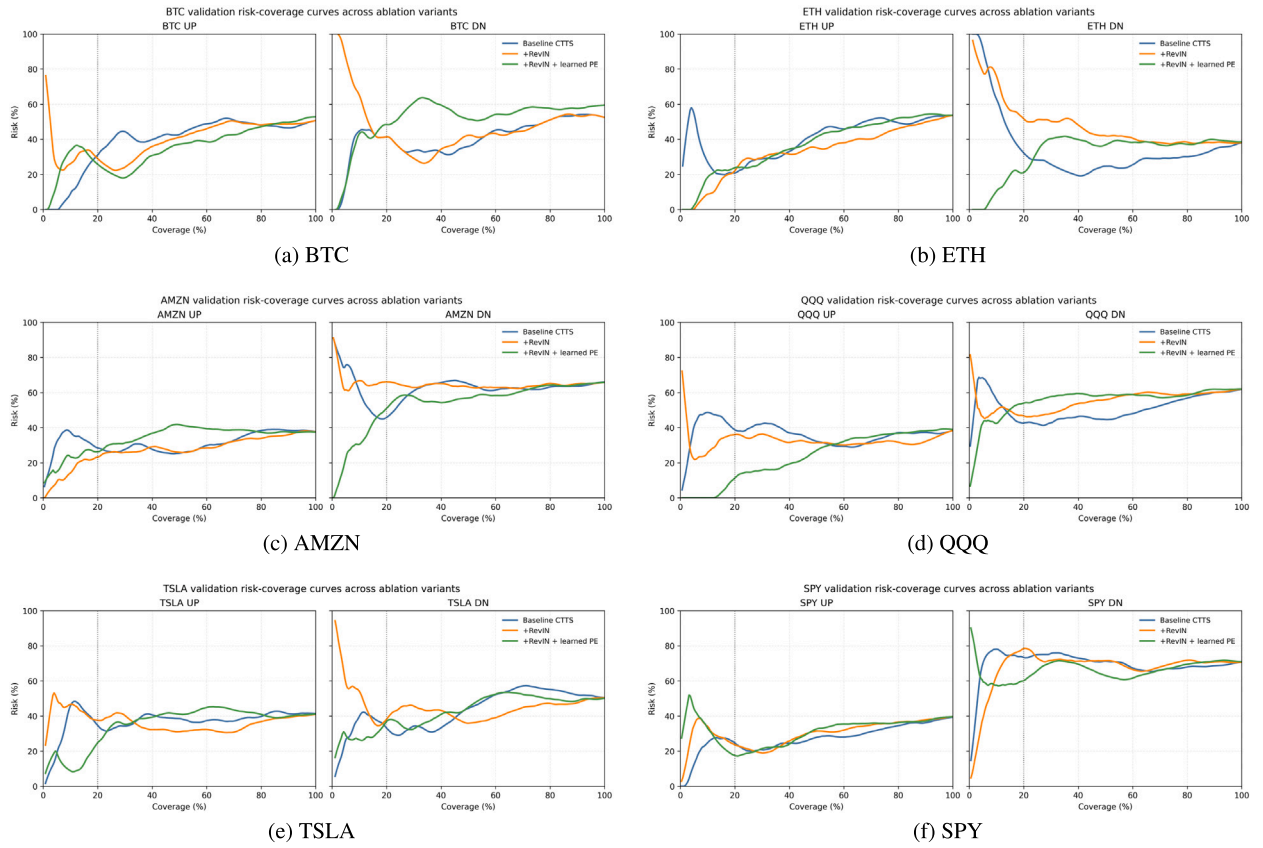


Fig. G.3. Validation risk–coverage curves for the three ablation variants (Baseline CTTS, +RevIN, +RevIN + Learned PE) across all six assets. Each panel reports UP and DOWN tasks for one asset. The vertical dotted line marks 20% coverage, the iso-coverage operating point reported in Table 3. Lower curves indicate a more favorable trade-off. The proposed variant (+RevIN + Learned PE, green) tends to achieve the lowest risk in the 20–40% coverage region on most UP tasks, with the largest advantages on QQQ and TSLA; the DOWN side shows greater variability across assets, consistent with the comparable aggregate Risk@20% Cov. numbers reported in Appendix Section C.

Appendix H. Self-attention weights

This appendix visualizes the self-attention weights of the CTTS reliability estimator ($M2$) across representative accept and reject decisions on the Bitcoin test set, shown in Fig. H.4.

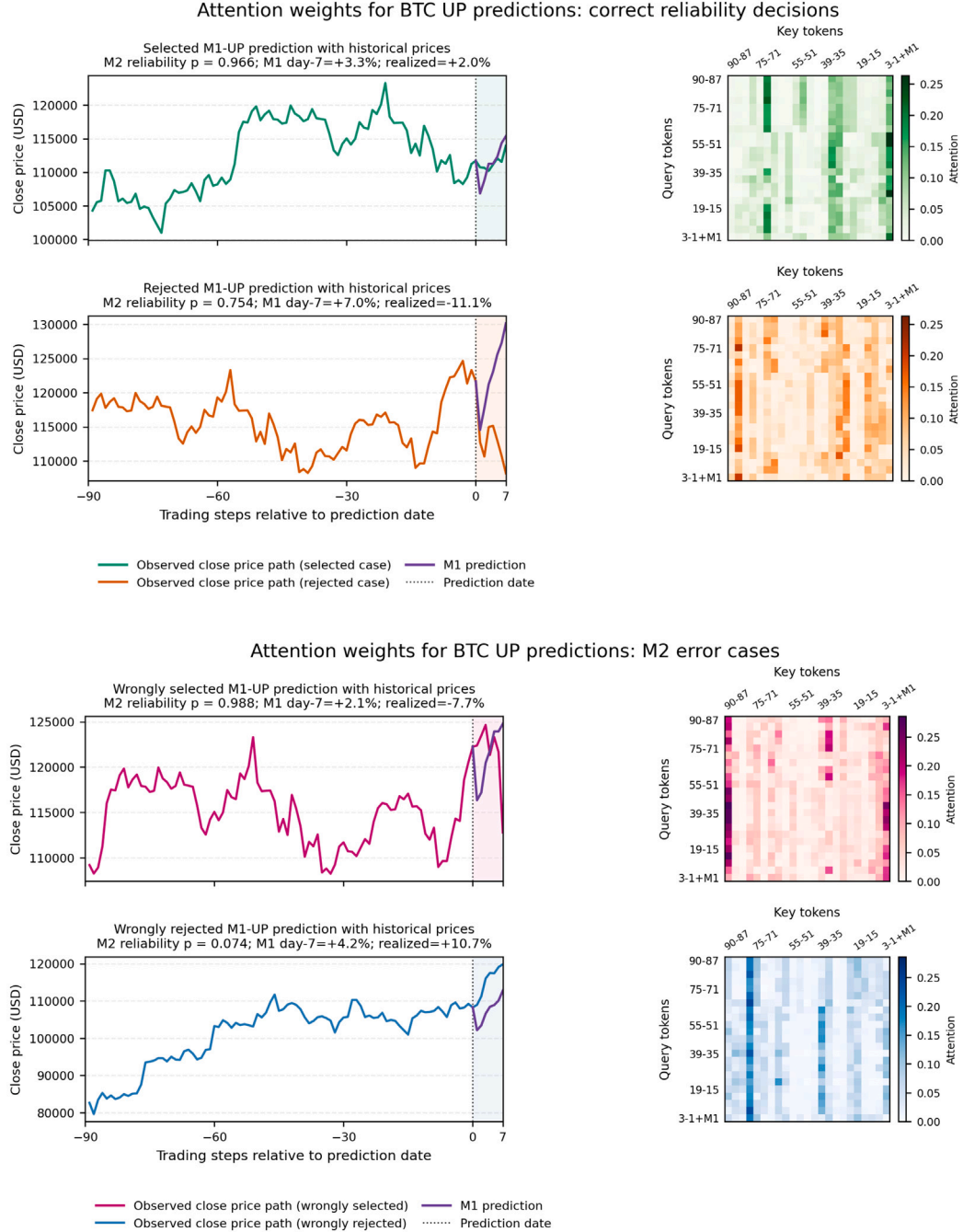


Fig. H.4. Self-attention weights from the CTTS reliability estimator ($M2$) for four representative $M2$ -decision $\times$ $M1$ -outcome cases on the Bitcoin test set. Row 1 (green): correct accept. Row 2 (Orange): correct reject. Row 3 (purple): incorrect accept (forecast turned out wrong). Row 4 (blue): incorrect reject (forecast turned out correct). Left panels show the 90-day lookback, the $M1$ prediction at $t = 0$ (purple), and the realized 7-step path; right panels show attention heatmaps averaged across layers, heads, and query positions. Accept decisions (Rows 1, 3) concentrate attention on recent tokens more than reject decisions (Rows 2, 4); Although the panels depict BTC examples, aggregating over $M2$'s accept/reject decisions across all assets gives a mean recent-token attention share of $\approx 42\%$ for accepts versus $\approx 28\%$ for rejects. Notably, this also holds for the wrong accept (Row 3), so attention alone does not separate correct from incorrect accepts.

Data availability

Data will be made available upon request.

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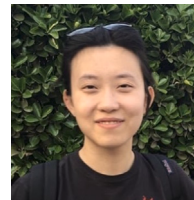
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